Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad Prov Original

Kurskod	Provkod	Tentamensdatum
MA080G	T102	2024-05-30
Kursnamn	Matematik GR (B), Kryptografi	
Provnamn	Skriftlig tentamen	
Ort	Sundsvall	
Termin		
Ämne		

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Tentamen
MA080G Kryptografi
30 May 2024

Mittuniversitetet
IMD
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Skrivtid: 5 timmar

Hjälpmedel: Miniräknare som ej är symbolhanterande. Ange märke och modell på din miniräknare på omslaget till tentamen. Inga böcker, anteckningar eller mobiltelefoner är tillåtna i tentamensrummet. This is a closed book examination. No books, notes or mobile telephones are allowed in the examination room. Electronic calculators may be used provided they cannot handle formulas. The make and model used must be specifed on the cover of your script.

Till alla uppgifter skall fullständiga lösningar lämnas. Resonemang och uträkningar får inte vara så knapphändiga att de blir svåra att följa. En uppgift per blad, skriv endast på en sida.

Betyg sätts efter hur väl lärandemålen är uppfyllda. Riktvärden för betygen är A 22p, B 18p, C 14p, D 10p och E 9p.

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1. Answer the following questions. **(4p)**
   (a) The message below is a ciphertext encrypted using a substitution cipher. What is the plaintext message?

   `C SJ WBSQ C SJ.`

   (b) Eve gets a message, and she knows it is a ciphertext encrypted using a Vigenère cipher. Using the Friedman's attack, she estimates that the length of the key is 4. She also has the letter frequencies by position modulo 4 (in the table on the next page). What is the key? Justify your answer.

2. How many subgroups does $\mathbb{Z}6$ have? Find two of them. **(2p)**

3. Calculate $\phi(540)$, where ϕ denotes Euler's totient function. **(2p)**
   Hint: Recall that $\phi(p^e) = p^e - p^{e-1}$.

4. Recall the two versions of Fermat's little theorem. **(3p)**
   (a) State one of them.
   (b) Use it to calculate $3^{18} \pmod{17}$.

5. Consider $p = 11$. **(3p)**
   (a) Using the Euler totient function, determine how many primitive roots $\mathbb{F}_p^\times$ has.
   (b) Find them all.

6. Your private key is $a = 9$. Use the *Diffie-Hellman Key exchange method* to create your public key, with $p = 11$ and $g = 6$. Use the fast exponentiation algorithm on your computation. **(2p)**

7. Alice and Bob are exchanging encrypted messages. The cryptosystem is ElGamal. Alice's public key is $p = 8779$, $g = 11$ and $h_A = 135$, while Bob's public key is $p = 8779$, $g = 11$ and $h_B = 2981$. **(4p)**
   (a) You are Bob, your private key is $b = 41$ and you want to encrypt the plaintext $m = 5347$ to send it to Alice. What is the ciphertext?
   (b) You are Alice, your private key is $a = 33$ and you receive the ciphertext $c = 1328$ from Bob. What is the plaintext?
8. The cryptosystem is RSA. Bob's public key is $n = 247$ and $e = 7$. **(4p)**
   (a) Use the Pollard ρ algorithm to factorize n. You must choose the polynomial function $f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ given by
   $$f(x) = x^2 + x + 1 \pmod{247}.$$
   (b) What is Bob's private key?

Table of letter frequency analysis for Question 1:

Letter	**1**	**2**	**3**	**4**
A	0	3	1	0
B	0	2	2	1
C	1	6	1	0
D	0	2	2	0
E	2	4	0	17
F	1	5	3	1
G	1	3	4	1
H	15	5	5	4
I	1	3	0	16
J	3	2	1	2
K	2	1	0	1
L	16	0	0	1
M	4	1	18	18
N	0	1	0	2
O	5	16	2	4
P	14	1	2	3
Q	1	2	17	3
R	3	0	1	5
S	2	14	1	4
T	15	1	5	1
U	5	5	16	5
V	0	3	4	5
W	2	13	1	1
X	4	1	1	2
Y	1	2	12	1
Z	2	4	1	0